VEKTOR

BY

INVESTPUPPY

VEKTOR RESEARCH SERIES · 2026

Technical Indicator Selection: Grid Search Methodology

A practitioner's guide to systematic parameter optimisation across six technical indicators — and how Sharpe-ranked selection produces consistent, instrument-specific trading signals.

For professional and institutional investors

IP-WP-TIS-260501-1.0

May 2026

ABSTRACT

Default indicator parameters — SMA 50/200, RSI 14 — were calibrated for the average instrument in average conditions. We do not use them. This paper explains why, and exactly what we use instead. Technical indicators are commonly applied with default or rule-of-thumb parameters — a practice that ignores the significant variation in optimal configurations across different instruments and market regimes. This paper presents the grid search methodology used by Vektor to identify the optimal indicator and parameter set for each individual holding: how the search space is defined, how performance is evaluated, how the winning configuration is selected, and how the process generalises across any listed equity market.

KEY TAKEAWAYS

— Default indicator parameters (e.g., SMA 50/200) are designed for the average — not for any specific instrument.

— A full grid search across the parameter space of six indicators identifies the configuration with the highest historical risk-adjusted return for each stock.

— Annualised Sharpe ratio is the primary selection criterion — it penalises high-volatility signals and rewards consistent, risk-adjusted performance.

— Three years of daily price data provides sufficient history for stable Sharpe estimation while remaining relevant to current market conditions.

1. WHY DEFAULT PARAMETERS FAIL

The problem with applying industry-standard indicator parameters to individual instruments.

The SMA 50/200 crossover — the so-called golden cross — is the most widely referenced technical signal in institutional and retail portfolio management. It is applied identically to every instrument in countless systematic strategies. The assumption embedded in this practice is that the same look-back period is equally relevant to a large-cap financial stock, a small-cap industrial, and a real estate trust in the same market. This assumption is demonstrably false.

Different instruments have different natural cycle lengths, different liquidity profiles, different volatility regimes, and different sensitivities to market-wide factors. An instrument that trends strongly over 30-day periods will not be well served by a 200-day slow average. An instrument with high intraday volatility may generate more false signals from an RSI configured for slower-moving stocks. The solution is not to find a better universal parameter — it is to abandon the premise of universality.

2. DEFINING THE SEARCH SPACE

How the parameter ranges for each indicator are specified and bounded.

For each of the six indicators, Vektor defines a bounded parameter space calibrated to the typical behaviour of listed equities with at least three years of price history. The ranges are set to cover the full spectrum of plausible configurations while excluding economically unreasonable extremes.

INDICATOR	PARAMETER	RANGE	STEP
SMA	Short period	10–30 days	1
SMA	Long period	40–80 days	1
EMA	Short period	10–30 days	1
EMA	Long period	40–80 days	1
MACD	Short EMA	5–20	1
MACD	Long EMA	20–50	1
MACD	Signal period	5–15	1
BB	SMA period	10–30 days	1
BB	Std deviations	1–3	0.5

RSI	Period	10–20 days	1
RSI	Upper threshold	60–80	5
RSI	Lower threshold	20–40	5
SO	Period	10–20 days	1
SO	%D smoothing	2–5	1

3. SHARPE RATIO AS SELECTION CRITERION

Why Sharpe ratio is the correct objective function — and what it captures that raw return does not.

Maximising raw return from a technical indicator backtest produces configurations that generate large gains but with commensurately large drawdowns — high-volatility signals that may perform well in trending markets but catastrophically in ranging ones. Minimising volatility alone produces overly conservative signals that generate few trades and limited performance. The Sharpe ratio balances both: it selects the configuration that produces the most return per unit of risk, consistently.

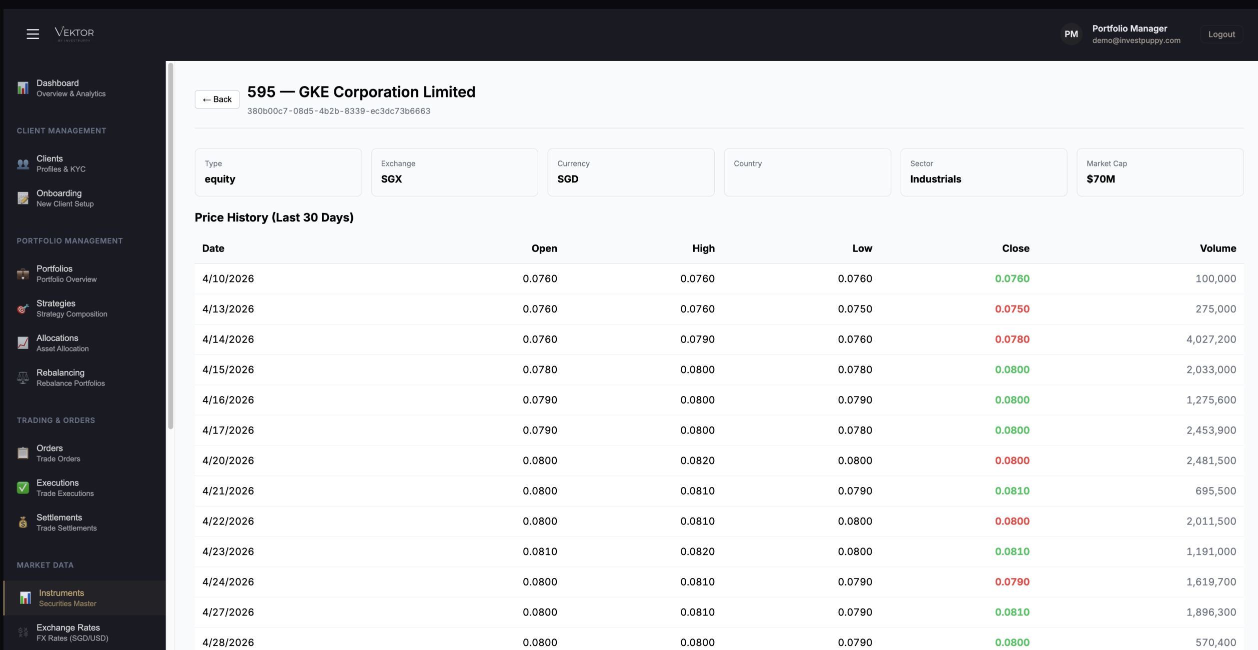
For each configuration tested, the annualised Sharpe ratio is calculated as: (annualised signal return – risk-free rate) / annualised standard deviation of signal returns. The risk-free rate used is the current national benchmark rate for the relevant market (e.g., Singapore 3-month T-bill rate for SGX mandates).

4. THREE YEARS OF DAILY DATA

Why the three-year look-back window is the correct balance between stability and relevance.

A shorter look-back period (e.g., one year) produces Sharpe estimates that are sensitive to recent market conditions — potentially selecting a configuration that performed well in a recent bull run but has no long-term validity. A longer look-back (e.g., ten years) incorporates market regimes that are no longer relevant to current conditions. Three years of daily data provides approximately 750 trading days — sufficient for statistically stable Sharpe estimation while remaining anchored to conditions that are reasonably representative of the current market.

PLATFORM INTERFACE



Date	Open	High	Low	Close	Volume
4/10/2026	0.0760	0.0760	0.0760	0.0760	100,000
4/13/2026	0.0760	0.0760	0.0750	0.0750	275,000
4/14/2026	0.0760	0.0790	0.0760	0.0780	4,027,200
4/15/2026	0.0780	0.0800	0.0780	0.0800	2,033,000
4/16/2026	0.0790	0.0800	0.0790	0.0800	1,275,600
4/17/2026	0.0790	0.0800	0.0780	0.0800	2,453,900
4/20/2026	0.0800	0.0820	0.0800	0.0800	2,481,500
4/21/2026	0.0800	0.0810	0.0790	0.0810	695,500
4/22/2026	0.0800	0.0810	0.0800	0.0800	2,011,500
4/23/2026	0.0810	0.0820	0.0800	0.0810	1,191,000
4/24/2026	0.0800	0.0810	0.0790	0.0790	1,619,700
4/27/2026	0.0800	0.0810	0.0790	0.0810	1,896,300
4/28/2026	0.0800	0.0800	0.0790	0.0800	570,400

Price history detail — daily OHLCV data auto-refreshed every morning at 8 AM SGT. Three years of history used for all signal backtests and grid searches.

5. OVERFITTING AND OUT-OF-SAMPLE VALIDITY

The most important question about any grid search: how do you know the optimised parameters work out of sample?

This is the correct question and it deserves a direct answer. A grid search that selects parameters purely on in-sample Sharpe ratio risks overfitting — selecting a configuration that performed well on historical data but degenerates on unseen data. Three design decisions explicitly address this risk.

Parameter range bounding. The search space is bounded by economically motivated ranges, not by whatever produces the highest in-sample Sharpe. This reduces effective degrees of freedom and limits overfitting scope.

Sharpe ratio as objective, not raw return. Maximising raw return on a backtest reliably produces overfitting. Sharpe ratio penalises volatility, selecting more stable, regime-robust configurations over those that fitted a specific trending period.

Daily re-optimisation against rolling data. The grid search re-runs against the most recent three years of data whenever a strategy is created or reviewed. This provides a continuous implicit out-of-sample test.

> Candour note: these design decisions reduce overfitting risk materially but do not eliminate it. No in-sample optimisation can guarantee out-of-sample performance. The XGBoost validation layer (WP-02) provides an additional real-time filter at the point of execution. The roadmap includes formal walk-forward validation as live track record develops.

6. CONCLUSION

Systematic parameter optimisation is not more complex than applying default parameters — it is simply more precise.

The grid search methodology ensures every instrument carries a trading signal calibrated to its own price behaviour — reproducible, transparent, and optimised for risk-adjusted performance. The overfitting risks inherent in parameter optimisation are addressed through bounded search spaces, Sharpe-based selection, and daily re-optimisation against rolling data.

This paper provides the technical detail behind Step 04 of the Vektor workflow. For XGBoost signal validation, see WP-02: Per-Instrument Signal Optimisation. For portfolio construction, see WP-01. All documents at investpuppy.com.